

# Myanmar Earthquake & Dam Risk Report

## 1. Executive Summary

This report covers 9,829 earthquake events in Myanmar from 1970-03-10 to 2026-08-28. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.038 with magnitude of completeness  $M_c=4.7$ .

**Total Events:** 9,829

**Date Range:** 1970-03-10 to 2026-08-28

**Max Magnitude:** M7.7

**Mean Depth:** 25.7 km

**b-value:** 1.038

**a-value:** 7.696

**$M_c$  (completeness):** 4.7

## 2. Earthquake Map

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## 3. Seismic Clustering Analysis

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## 4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

**Critical Risk:** 44 dams

**High Risk:** 116 dams

**Moderate Risk:** 45 dams

**Low Risk:** 49 dams

### Top 30 Highest Risk Dams:

| Name                   | River          | PGA(g) | Dist    | Fault   | Risk     | Grade    |  |  |  |
|------------------------|----------------|--------|---------|---------|----------|----------|--|--|--|
| Dam near 21.74N 95.85E |                | nan    | 0.2718  | 7.4 km  | 10.0     | Critical |  |  |  |
| Dam near 22.09N 95.86E |                | nan    | 0.2995  | 5.7 km  | 10.0     | Critical |  |  |  |
| Dam near 21.77N 95.89E |                | nan    | 0.3257  | 2.7 km  | 10.0     | Critical |  |  |  |
| Dam near 21.47N 95.87E |                | nan    | 0.2815  | 6.0 km  | 9.7      | Critical |  |  |  |
| Dam near 20.85N 96.03E |                | nan    | 0.4813  | 8.3 km  | 8.8      | Critical |  |  |  |
| Meikhtila              | nan            | 0.2412 | 8.2 km  | 8.8     | Critical |          |  |  |  |
| Dam near 21.34N 95.81E |                | nan    | 0.2263  | 12.2 km | 8.8      | Critical |  |  |  |
| Natkar                 | nan            | 0.2443 | 7.9 km  | 8.7     | Critical |          |  |  |  |
| Dam near 21.37N 95.80E |                | nan    | 0.2207  | 13.4 km | 8.7      | Critical |  |  |  |
| Dam near 20.72N 95.90E |                | nan    | 0.2665  | 5.6 km  | 8.7      | Critical |  |  |  |
| Dam near 21.04N 95.83E |                | nan    | 0.2270  | 10.9 km | 8.7      | Critical |  |  |  |
| Dam near 21.05N 95.83E |                | nan    | 0.2265  | 11.0 km | 8.7      | Critical |  |  |  |
| Dam near 20.47N 96.08E |                | nan    | 0.5000  | 9.9 km  | 8.6      | Critical |  |  |  |
| Dam near 20.60N 95.90E |                | nan    | 0.2464  | 7.4 km  | 8.6      | Critical |  |  |  |
| Taungnyo               | Taungnyo       | 0.3054 | 1.2 km  | 8.6     | Critical |          |  |  |  |
| Nat Thar Taw           | Nat Thar Taw   | 0.1903 | 20.2 km | 8.5     | Critical |          |  |  |  |
| Myotha                 | Ku Hta         | 0.1849 | 20.6 km | 8.5     | Critical |          |  |  |  |
| Thaphan Chaung         | Thaphan Chaung | 0.2197 | 11.5 km | 8.2     | Critical |          |  |  |  |
| Zeedaw                 | Nyaung Oat     | 0.2075 | 15.3 km | 8.2     | Critical |          |  |  |  |
| Dam near 17.55N 96.37E |                | nan    | 0.1022  | 9.6 km  | 8.2      | Critical |  |  |  |
| Male-Nattaung          | Male-Nattaung  | 0.2360 | 18.6 km | 8.1     | Critical |          |  |  |  |
| Dam near 20.73N 95.83E |                | nan    | 0.2110  | 13.6 km | 7.9      | Critical |  |  |  |
| Nyaunggone             | nan            | 0.2104 | 13.7 km | 7.9     | Critical |          |  |  |  |
| Paunggadaw             | Paunggadaw     | 0.1488 | 33.9 km | 7.9     | Critical |          |  |  |  |
| Kun Chaung             | Kun Chaung     | 0.4644 | 0.9 km  | 7.7     | Critical |          |  |  |  |
| Kintat                 | Mu             | 0.1669 | 21.0 km | 7.7     | Critical |          |  |  |  |
| North Pinle            | Pinle          | 0.1427 | 36.1 km | 7.7     | Critical |          |  |  |  |
| Phyu Chaung            | Pyu Chaung     | 0.4707 | 1.3 km  | 7.6     | Critical |          |  |  |  |
| South Pinle            | Pinle          | 0.1383 | 38.1 km | 7.6     | Critical |          |  |  |  |
| Kabaung                | Kabaung Chaung | 0.3335 | 9.2 km  | 7.6     | Critical |          |  |  |  |

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## 5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

**K (productivity):** 34.92

**c (time offset):** 16.408 hours

**p (decay exponent):** 0.772

**Expected M $\geq$ 3 (7d):** 4

**Expected M $\geq$ 3 (30d):** 17

**Expected M $\geq$ 3 (90d):** 49

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

## 6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit